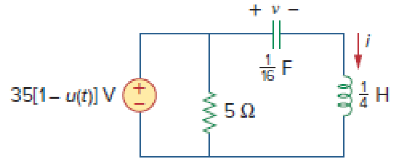
Chapter 16, Problem 29P

Problem

Calculate $i(t)$ for $t > 0$ in the circuit in Fig. 16.52.

Figure 16.52:



Step-by-step solution

Step 1 of 6

Refer to Figure 16.52 from the textbook.

The voltage source $35[1-u(t)] = 35\text{ V}$ for $t<0$.

Consider that the circuit is in steady state for $t<0$. The capacitor acts as an open circuit and the inductor acts as a short circuit in steady state.

Modify and redraw the circuit diagram for $t<0$.

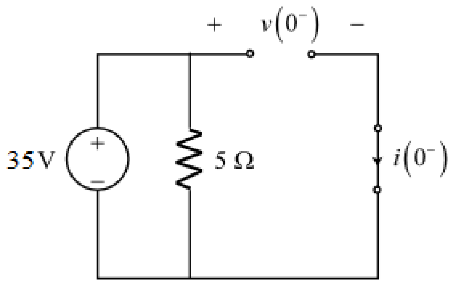


Figure 1

Step 2 of 6

Calculate the current $i(0^-)$ from the circuit in Figure 1.

$$i(0^-) = 0\text{ A}$$

Inductor does not allow sudden changes in the current flowing through it. Therefore,

$$\begin{aligned} i(0^-) &= i(0^+) \\ &= i(0) \\ &= 0\text{ A} \end{aligned}$$

Calculate the voltage $v(0^-)$ from the circuit in Figure 1.

$$v(0^-) = 35\text{ V}$$

Capacitor does not allow sudden changes in the voltage across it. Therefore,

$$\begin{aligned} v(0^-) &= v(0^+) \\ &= v(0) \\ &= 35\text{ V} \end{aligned}$$

Step 3 of 6

The voltage source $35\left[1-u(t)\right]=0$ (short circuit) for $t>0$.

Modify and redraw the circuit diagram for $t>0$.

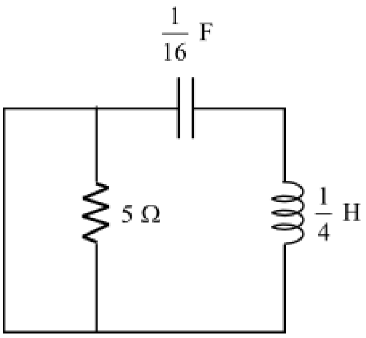


Figure 2

Step 4 of 6

Simplify the circuit in Figure 2.

$R=0$

Represent the circuit in *s*-domain including the initial values.

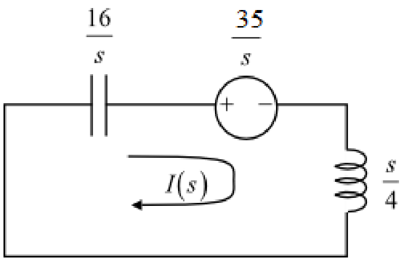


Figure 3

Comments (1)

Anonymous

why can we set R=0?

Step 5 of 6

Apply Kirchhoff's voltage law to the circuit in Figure 3.

$$\frac{16}{s}I(s)+\frac{35}{s}+\frac{s}{4}I(s)=0$$

$$I(s)\left[\frac{16}{s}+\frac{s}{4}\right]=-\frac{35}{s}$$

$$I(s)\left[64+s^2\right]=-\frac{35}{s}(4s)$$

$$I(s)=\frac{-140}{s^2+8^2}$$

$$=-17.5\left(\frac{8}{s^2+8^2}\right)$$

Step 6 of 6

Find the inverse Laplace transform to find the current $i(t)$.

Use the following properties of the Laplace transform:

$$\sin \omega t \leftrightarrow \frac{\omega}{s^2+\omega^2}$$

Find the inverse Laplace transform of the function $I(s)$.

$$i(t) = -17.5L^{-1}\left\{\frac{8}{s^2+8^2}\right\}$$

$$\begin{aligned} i(t) &= -17.5\sin 8tu(t) \\ &= 17.5[-\sin 8t]u(t) \\ &= 17.5\cos(8t+90°)u(t) \text{ A} \end{aligned}$$

Thus, the current $i(t)$ for $t>0$ is $\boxed{i(t) = 17.5\cos(8t+90°)u(t) \text{ A}}$.